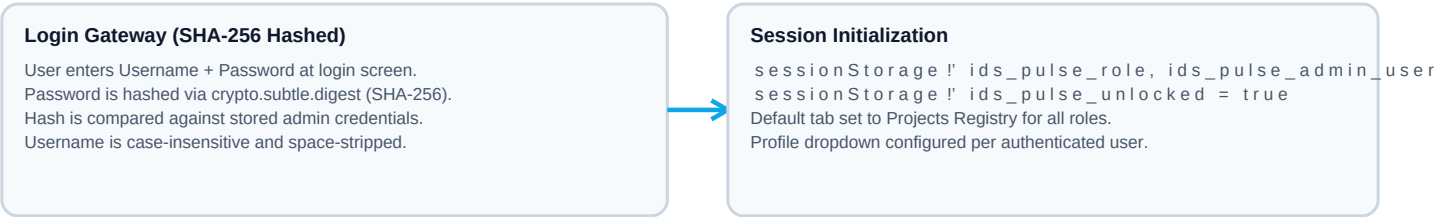


1. USER AUTHENTICATION FLOW

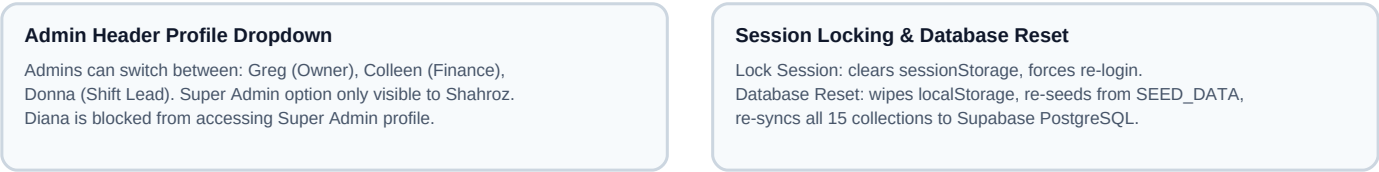


2. USER ROLES & ACCESS MATRIX

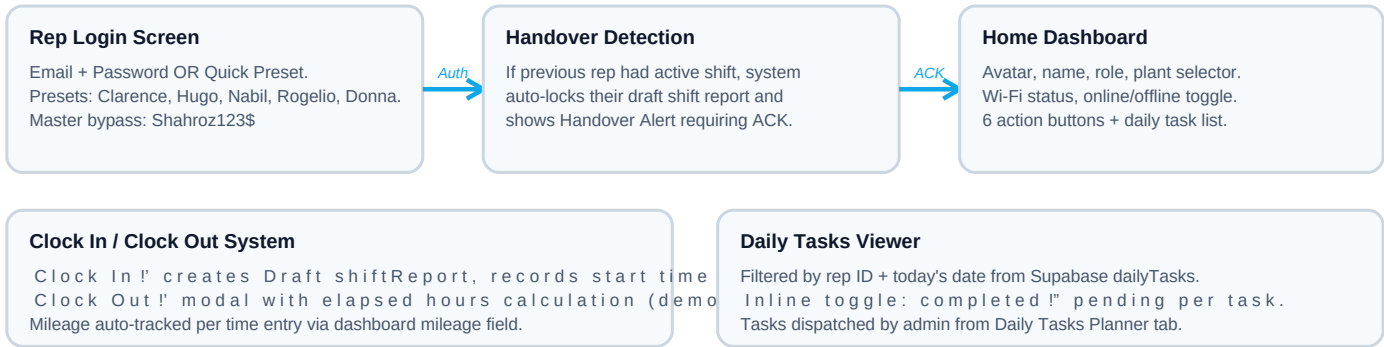
| Super Admin | Owner                             | Accountant | Shift Lead | Field QRE                      | Customer                            |
|-------------|-----------------------------------|------------|------------|--------------------------------|-------------------------------------|
| Shahroz     | Diana, Greg, Monica, Iris, Miriam | Colleen    | Donna      | Hugo, Nabil, Rogelio, Clarence | AutoKabel, Magna, Hutchinson, Brose |

| Feature / Tab             | Super Admin | Owner | Accountant | Lead | QRE      | Customer       |
|---------------------------|-------------|-------|------------|------|----------|----------------|
| Incidents & Defect Feed   | ✔           | ✔     | —          | ✔    | —        | —              |
| Visual Defect Heatmap     | ✔           | ✔     | —          | ✔    | —        | —              |
| Daily Tasks Planner       | ✔           | ✔     | —          | ✔    | —        | —              |
| Shift Summaries & Publish | ✔           | ✔     | —          | ✔    | —        | Published Only |
| Suppliers Directory       | ✔           | ✔     | —          | —    | —        | —              |
| Timesheets & Mileage      | ✔           | ✔     | ✔          | —    | Own Only | Own Supplier   |
| Invoice & Billing         | ✔           | ✔     | ✔          | —    | —        | —              |
| Rework Logs Feed          | ✔           | ✔     | —          | ✔    | —        | —              |
| Email Logs                | ✔           | ✔     | —          | —    | —        | —              |
| User Directory            | ✔           | ✔     | —          | —    | —        | —              |
| Projects Registry         | ✔           | ✔     | —          | —    | —        | —              |
| Setup & Onboarding (CRUD) | ✔           | ✔     | —          | —    | —        | —              |
| System Events Logger      | ✔           | —     | —          | —    | —        | —              |
| Launch Roadmap            | ✔           | —     | —          | —    | —        | —              |
| Customer Portal           | —           | —     | —          | —    | —        | ✔              |
| Pulse AI Chatbot          | ✔           | ✔     | ✔          | ✔    | ✔        | —              |

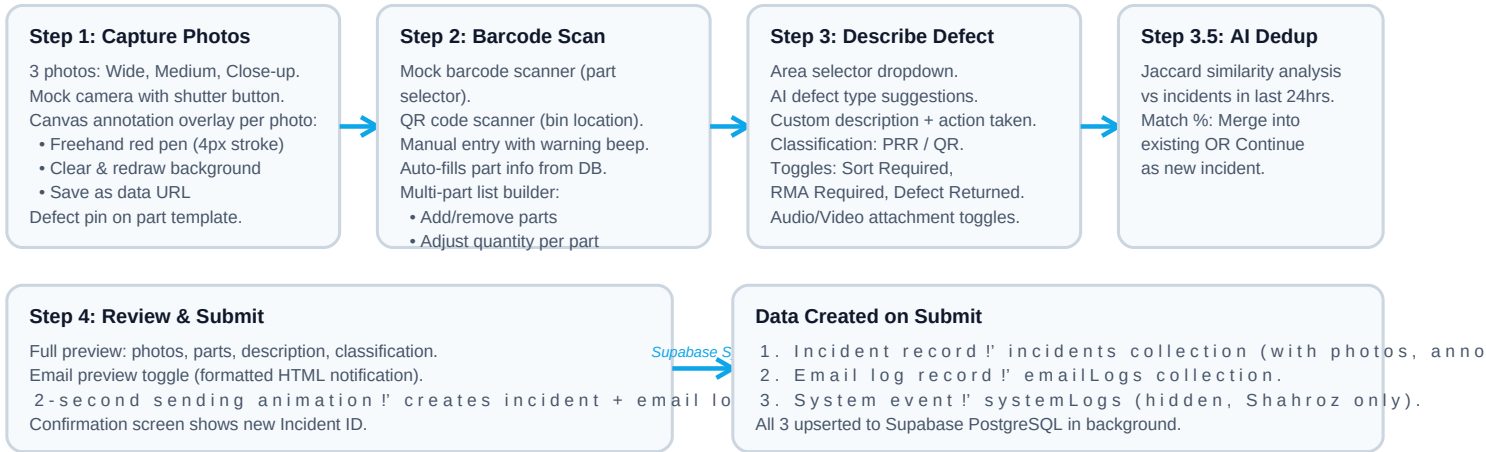
3. PROFILE SWITCHER & SESSION SECURITY



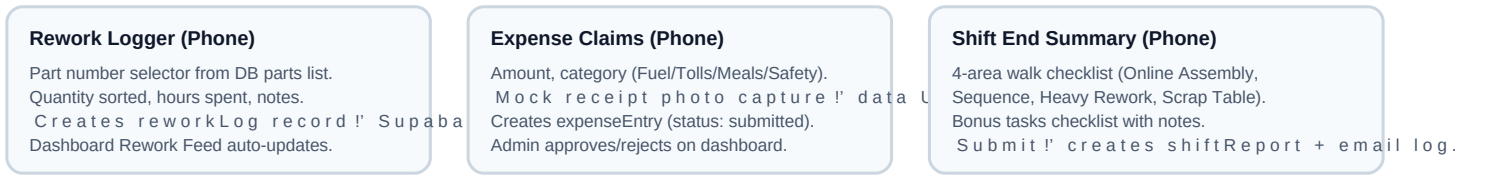
4. PHONE SIMULATOR — COMPLETE FIELD REP WORKFLOW



5. INCIDENT REPORT WIZARD (4-STEP + AI DUPLICATE CHECK)



6. REWORK LOGGER, EXPENSE CLAIMS & SHIFT SUMMARY



7. ADMIN DASHBOARD TABS — OPERATIONS & QUALITY MANAGEMENT

**TAB: Incidents & Defect Feed**

Full list of incident reports with search, supplier filter, status filter.  
Detail drawer (slide-out): photo gallery, annotations, parts list, coordinates.  
Status toggle: Open | Closed (logged to system events)  
EXPORTS: Individual incident audit PDF (with logo, confidentiality badge, watermark). Browser Print dialog. Resend supplier email notification.

**TAB: Visual Defect Heatmap Matrix**

Interactive defect location heatmap on part template image.  
Part selector dropdown (by part number from DB).  
Time scrubber to visualize defect clustering over time.  
Hover tooltips showing individual incident details.  
Dot opacity based on recency of defect occurrence.

**TAB: Daily Tasks Planner**

Date-scoped task assignment per representative.  
Rep filter dropdown to view specific rep's tasks.  
Create new tasks: assign instruction, rep, date.  
Toggle task status: completed | pending.  
Tasks pushed live to Phone Simulator for rep.

**TAB: Shift Summaries Log**

List of all shift walkthrough reports from reps.  
Detail drawer: areas walked, bonus tasks, incidents count.  
Publish to Customer: changes status! 'published', making report visible in Customer Portal.  
EXPORTS: Shift walkthrough audit PDF, browser print.

**TAB: Suppliers Directory**

Grid of supplier cards: name, invoice schedule, allotted hours, contacts (name, email, role), plants served list.  
EXPORTS: Full supplier contacts directory PDF, browser print.

**TAB: Rework Logs Feed**

All rework/sort activities with date filtering.  
Detail drawer per individual rework log.  
EXPORTS: Feed-level PDF + individual rework audit PDF.  
Browser print for both feed and individual reports.

**TAB: Email Logs**

Full email delivery audit trail.  
Detail view: to, cc, subject, HTML body, delivery status, timestamp.

**TAB: User Directory**

Grid of user cards: name, email, phone, role, avatar, pay currency.  
All registered reps, leads, owners, and accountants.

8. PULSE AI — CONVERSATIONAL DATABASE AUDITOR

**AI Chat Console (Left Panel)**

Natural language chat interface with message history.  
Role-specific welcome messages and quick prompt chips.  
Admin: 'Audit Database', 'Scan Duplicates', 'Export Excel/CSV'.  
Accountant: 'Audit Timesheets', 'Export Excel/CSV'.  
Lead: 'Audit Quality Logs', 'Download Quality PDF'.  
QRE: 'Check my hours', 'What tasks do I have today?'.

**AI Audit Engine (Right Panel)**

Run Scan button triggers comprehensive data audit:

- Flags duplicate defect incidents (same part + area).
- Detects timesheet anomalies (negative hrs, >24hr entries).
- Identifies receipt mismatches in expense claims.
- Checks daily hour limit breaches.

Triggers: Excel, CSV, PDF exports via chat commands.

9. TIMESHEETS, INVOICING & FINANCIAL OPERATIONS

Sub-Tab: Log Hours & Expenses

Log Hours: rep, supplier, date, hours, mileage (km), notes ! timeEntries.  
Log Expense: rep, supplier, date, category, amount, notes ! expenseEntries.  
Both sync to Supabase immediately.

Sub-Tab: Invoice Generator

Supplier selector, currency filter (CAD/USD).  
Calculates: hours × billing rate + mileage × \$0.73/km + expenses.  
Mark entries as invoiced.  
PDF invoice + QuickBooks CSV export.

Sub-Tab: Payroll Approvals

Expense approval queue (approve/reject).  
Extra hours admin final approval queue (after customer approval).  
Auto-creates timeEntry on approval.

Sub-Tab: Rates & Configuration + Admin CRUD

Rate override matrix: custom billing/pay rates per rep + supplier.  
CRUD Sub-Tab: Customers ! register new client/supplier.  
CRUD Sub-Tab: Locations ! map new plant, link to supplier.  
CRUD Sub-Tab: Reps ! onboard new representative (national currency).

Sub-Tab: Bulk Entry + Quick Add Modals

Quick Add Rep: inline modal popup, auto-selects in registry.  
Quick Add Client: inline modal popup with invoice schedule.  
Quick Add Plant: inline modal popup with address + supplier link.  
All modals write directly to Supabase via saveEntity().

10. EXTRA HOURS — 3-STAGE APPROVAL WORKFLOW

Stage 1: Rep Files

Rep submits extra hours request with: supplier, plant, date, hours, reason.  
Status ! pending\_customer

Notify

Stage 2: Customer Review

Customer sees request in their Portal approval queue.  
Can approve or reject with comment. Status ! pending\_admin

Escalate

Stage 3: Admin Final

Admin reviews in Payroll tab.  
Approve ! auto-creates new timeEntry record in timesheets.  
Reject ! status: rejected\_by\_admin

Result

Final Outcome

Approved: timeEntry created with '[APPROVED EXTRA HOURS]' note. Appears in timesheets.  
History trail with comments.

11. EXPENSE CLAIMS — 2-STAGE APPROVAL WORKFLOW

Stage 1: Rep Submits Expense

Rep files expense via Phone or Dashboard.  
Includes: amount, category, receipt photo, notes.  
Status ! submitted. Synced to Supabase.

Queue

Stage 2: Admin Approves/Rejects

Admin reviews in Payroll & Expense Approvals.  
View receipt photo (lightbox modal).  
Approve ! status: approved. Reject ! status: rejected.

Impact

Impact on Billing

Approved expenses included in client invoice calculations and payroll reports.  
Receipt photos embedded in audit PDFs.

12. CUSTOMER PORTAL (CLIENT-FACING DASHBOARD)

Location & QRE Assignment Grid

Shows customer's plants with assigned QRE rep.  
Unbilled hours progress bar per location.  
Weekly allotted vs actual hours comparison.

Extra Hours Approval Queue

Customer can approve/reject overtime requests filed by reps working at their plants.  
Comment field for approval/rejection rationale.  
Approved requests escalate to Admin.

Published Reports & Hours Audit

View published shift walkthrough reports.  
Weekly hours audit table filtered to customer's supplier\_id only.  
Billing rates are hidden from customer.

13. DATABASE ARCHITECTURE — 15 ENTITY COLLECTIONS

| Collection         | Description                                       |
|--------------------|---------------------------------------------------|
| users              | Reps, leads, owners, accountants (8 seed)         |
| rates              | Billing/pay rate overrides per rep+supplier+plant |
| plants             | OEM assembly plant locations (6 seed)             |
| suppliers          | Client/supplier companies with contacts (5 seed)  |
| parts              | Part number registry for barcode scanning         |
| incidents          | Defect incident reports with photos & annotations |
| shiftReports       | Shift walkthrough audit summaries                 |
| reworkLogs         | Rework/sort activity logs                         |
| timeEntries        | Hours + mileage per rep per day                   |
| emailLogs          | Email delivery audit trail                        |
| dailyTasks         | Assigned daily tasks per rep                      |
| expenseEntries     | Expense claims with receipt photos                |
| projects           | Project registry with billing/pay/currency        |
| extraHoursRequests | Overtime approval workflow items                  |
| systemLogs         | System event audit trail (Shahroz only)           |

14. SUPABASE SYNC MECHANISM (BROWSER LOCALSTORAGE !' POSTGRESQ

1

Initialize & Pull

On app load, initializeDB() seeds local cache from SEED\_DATA, then calls syncWithSupabase() after 100ms delay. For each of 15 collections, pulls from Supabase. If Supabase has data !' overwrites local. If empty !' seeds Supabase from local.

2

Push on Write

Every saveEntity() call writes to Browser LocalStorage immediately (zero latency) and simultaneously fires supabase.from(type).upsert(entity) in background. Delete operations call supabase.from().delete().eq('id', entityId).

3

Cross-Component Reactivity

After every write, a custom DOM event 'ids\_pulse\_db\_update' is dispatched. Both PhoneSimulator and WebDashboard listen for this event and re-read their local state from the updated cache.

4

Real-Time Notifications

Dashboard detects new data via dbUpdateTrigger prop. Plays dual-tone chime (Web Audio API sine wave). Shows toast notification banner for incidents, rework logs, shift reports, and expense claims.

5

Role-Based Data Sanitization

getEntities() applies per-role data filtering. Reps see only their own records. Customers see only their supplier's data with billing rates stripped. Admins get full unrestricted access.

15. COMPLETE REPORT & EXPORT CATALOG

PDF REPORTS (7 Types)

| Report Name            | Output Filename                         | Generated From                 |
|------------------------|-----------------------------------------|--------------------------------|
| Incident Audit PDF     | IDS_Pulse_Audit_{PN}_{incld}.pdf        | Incidents tab — per incident   |
| Shift Walkthrough PDF  | IDS_Shift_Walkthrough_{date}_{srid}.pdf | Shift Logs tab — per report    |
| Supplier Directory PDF | IDS_Supplier_Contacts_Directory.pdf     | Suppliers tab — full directory |
| Timesheet Payroll PDF  | IDS_Timesheets_Audit_Report.pdf         | Timesheets tab — full payroll  |
| Rework Feed PDF        | IDS_Rework_Audit_Feed_{date}.pdf        | Rework Logs tab — all rework   |
| Individual Rework PDF  | IDS_Rework_Audit_{rwid}.pdf             | Rework Logs tab — per rework   |
| Client Invoice PDF     | Invoice_{ClientName}_{timestamp}.pdf    | Invoice Generator sub-tab      |

SPREADSHEET EXPORTS (3 Types)

| Export Name          | Output Filename                    | Generated From                    |
|----------------------|------------------------------------|-----------------------------------|
| QuickBooks CSV       | QuickBooks_Export_{supplier}.csv   | Invoice Generator — per client    |
| Payroll CSV          | IDS_Timesheets_Payroll_{date}.csv  | Timesheets tab — full payroll     |
| Styled Excel (.xlsx) | IDS_Timesheets_Payroll_{date}.xlsx | Timesheets tab — ExcelJS workbook |

BROWSER PRINT REPORTS (5 Types)

- 1. Incident Audit Print — styled HTML from Incidents tab
- 2. Shift Walkthrough Print — styled HTML from Shift Logs tab
- 3. Supplier Directory Print — styled HTML from Suppliers tab
- 4. Timesheet Payroll Print — styled HTML from Timesheets tab
- 5. Rework Feed Print / Individual Rework Print — styled HTML from Rework Logs tab

16. HIDDEN SYSTEM EVENTS LOGGER (SUPER ADMIN ONLY)

Logged Event Categories

- auth: Login authentication events.
- system: CRUD operations (create customer, location, rep, save/delete rate overrides, quick-add actions).
- shift: Clock in/out, shift report submit, publish to customer.
- incident: New incident creation, status updates (open/close).
- rework: Rework log submissions.
- payroll: Overtime approval, expense approval, CSV/Excel/PDF exports.

Security & Access

System Events Logs tab is HIDDEN from all users except Shahroz. Sidebar button wrapped in userRole === 'shahroz' guard. Tab content double-guarded: activeTab + userRole check. Each event log records: timestamp, acting user (from sessionStorage), category, action type, and detail string. Clear console button resets log to single 'cleared' entry. All events synced to Supabase systemLogs table.

17. PROJECTS REGISTRY & QUICK ADD SYSTEM

Projects Registry Tab

List of registered projects: rep, client, plant, rates, currency, status.  
Create new project form with billing rate, pay rate, currency per project.  
Projects used by rate resolver — project rates take priority over rate overrides.

Quick Add Modals (Inline Popups)

Quick Add Rep: name, email, phone, currency!' auto-selects  
Quick Add Client: name, invoice schedule!' saves to supplier  
Quick Add Plant: name, address, linked supplier!' saves to p  
Available from Projects Registry tab and Rates Config sub-tab.